

Unlocking Flexibility Potential through DVFS-controlled Data Center Clusters in Unit Commitment

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Abstract—Data center clusters (DCCs) have become recognized as significant energy consumers in electrical power system operations. Workload reallocation processes for multiple DCCs, implemented by dynamic voltage and frequency scaling (DVFS) control schemes, can provide flexibility potential capabilities and contribute to maintaining the security of power systems operations through Unit Commitment (UC). To achieve this, The flexibility potential of workload reallocation hosted by multiple DCCs is analyzed. Building on this analysis, an interval-based workload reallocation strategy is extended to depict flexibility potential from workload response while avoiding service-time latency. Then, an innovation set of physical security constraints representing DVFS-controlled DCCs is developed, and related special-ordered sets-of-type-1 (SOS1) linearized constraints are derived through the McCormick Envelopes method. Furthermore, these intractable constraints are incorporated into UC, and an enhanced UC model, which considers the flexibility potential offered by DVFS-controlled DCCs, is developed. Several experiments are conducted on the IEEE 6 buses test systems to assess the validity and scalability of the proposed models.

Index Terms—Data center clusters, flexibility capability, workload reallocation, dynamic voltage and frequency scaling, unit commitment, McCormick envelopes

NOMENCLATURE

Index and Sets

$b \in \mathcal{B}$	Index and Sets of Buses
$d \in \mathcal{D}$	Index and Sets of conventional demands
$e \in \mathcal{E}$	Index and Sets of battery energy storage systems(BESS)
$g \in \mathcal{G}$	Index and Sets of conventional generators
$i \in \mathcal{I}$	Index and Sets of DCC servers
$l \in \mathcal{L}$	Index and Sets of transmission lines
$s \in \mathcal{S}$	Index and Sets of scenarios
$t \in \mathcal{T}$	Index and Sets of time periods
$w \in \mathcal{W}$	Index and Sets of wind farms

Parameters

γ_e^{ch}	Charging ramping capability of BESS e
γ_e^{dis}	Discharging ramping capability of BESS e
$\gamma_g^{su/sd}$	Shut-up/shut-down ramping capabilities of conventional generator g
$\gamma_g^{up/dw}$	Up/down-forward ramping capabilities of conventional generator g
$\iota_{l,b}$	Generating shifting distribution factor between transmission line l and bus b
$\lambda_{dcc,i,t}^{ini}$	Initial load-rate of DCC i at period t
$C_{dcc,i}$	Effective capacitance of DCC i
C_{sd}	Shut-down cost of conventional generator g
$C_{shed,w}$	Power curtailment cost of wind farm w
C_{su}	Shut-up cost of conventional generator g
$C_{voll,d}$	Value of lost load d

$i_{leak,i}$	Leakage current of DCC i
p_l	Power of transmission line l
$p_{dcc,i,t}^{dyn}$	Dynamic workload of DCC i at period t
$p_{dcc,i,t}^{idle}$	Static workload of DCC i at period t
T_{tol}	Latency threshold for workload processing delays
Variables	
$\lambda_{dcc,i,t}$	Workload rate of DCC i at period t
ω_g^{dt}	Min shut-down duration of conventional generator g
ω_g^{ut}	Min shut-up duration of conventional generator g
$\phi_{dcc,i,t}^{dvs}$	Power regulation of DCC i via DVFS technique
$\Delta\lambda_{dcc,i,t}$	Adjusted load-rate of DCC i at period t
$\Delta p_{d,t,s}$	Shedded load d at period t under scenario s
$\Delta p_{dcc,i,t}^{bal}$	Shifted workload of DCC i at period t
$\Delta p_{w,t,s}$	Abandoned output of wind farm w at period t under scenario s
$\varepsilon_{e,t,s}^{ch}$	Charging state decision of BESS e at period t under scenario s
$\varepsilon_{e,t,s}^{dis}$	Discharging state decision of BESS e at period t under scenario s
$f_{dcc,i,t}$	Operational frequency of DCC i at period t
$p_{add,i,t,s}$	workload of DCC i at period t under scenario s
$p_{d,t,s}$	Power of demand d at period t under scenario s
$p_{dcc,i,t}$	Workload consumption of DCC i at period t
$p_{e,t,s}^{ch}$	Charging power of BESS e at period t under scenario s
$p_{e,t,s}^{dis}$	Discharging power of BESS e at period t under scenario s
$p_{g,t,s}$	Output of conventional generator g at period t under scenario s
$p_{w,t,s}$	Stochastic output of wind farm w at period t under scenario s
$u_{g,t}$	Shut-down decision of conventional generator g at period t
$v_{dcc,i,t}$	Supply voltage of DCC i at period t
$v_{g,t}$	Shut-up decision of conventional generator g at period t
$x_{g,t}$	Operation state of conventional generator g at period t

I. INTRODUCTION

AS the increase in various artificial intelligence technologies and demand energy consumption, the operation of the electrical power system is becoming diverse and uncontrollable. It faces challenges in providing sufficient flexible adjustment resources against emergence events [1]. In such circumstances, additional flexible mechanisms, such as demand response (DR), virtual power plant (VPP), and battery energy

storage systems (BESS), need to be prioritized. Nevertheless, as a significant energy consumer, DCCs have not been fully utilized in the flexibility potential of power system operations. To this end, it is essential to investigate the flexible adjustment characteristic of DCCs and estimate their flexibility potential for the economical and reliable operation of electrical power systems.

Recent research has reported considerable advancements in understanding the potential of DCCs to enhance flexibility within electrical power systems. Specially, Ref. [2] carried out a quality-of-service-based cost function to describe the workload shifting and server utilization of DCCs. This study is further extended a distributed dispatch framework to integrate DCCs flexibility. Ref. [3] addressed the optimal demand management challenge among DCC servers through a well-designed mix-integer programming model. Ref. [4] proposed two optimization models to estimate the demand response capacities of DCCs. Ref. [5] explored the coupling relationship between DCCs and hydrogen resources in microgrid systems and subsequently carried out a data-driven rolling optimization model so as to ensure the optimal operation of DCCs and hydrogen storage systems. Ref. [6] analyses the flexibility potential of DCCs while concurrently reducing carbon emissions, culminating in an emission-aware coordinated dispatch model for multi-regional microgrids. While these studies highlight the flexibility of DCCs within power systems, the intricate physical security characteristics associated with DCCs were not detailed.

The workloads in DCCs can be broadly categorized into three types, [3], *i) compute-intensive workloads*, which include high-performance computing, machine learning model training, and commercial application services. *ii) storage-intensive workloads*, which are employed for such activities such as regular file backups, archival systems, and relational database applications. *iii) network-intensive workloads*, these jobs encompass activities like website hosting, content delivery networks, and network traffic management. Notably, compute-intensive and storage-intensive workloads dominate DCCs, collectively accounting for approximately 70% of the total workload amount, [7]. Moreover, the growing demand for high-performance computing and data-driven techniques has been contributing to this dominance. In contrast, network-intensive workloads are typically less predictable and highly time-sensitive, requiring immediate processing to ensure seamless operations. Since compute-intensive and storage-intensive workloads are predictable and flexible, they can be effectively shifted and reallocations to decrease their peak requirements of power consumption. Techniques like DVFS allow for shifting workloads or responding to delays, leading to improved spatial-temporal workload balancing and overall energy management.

In addition to possible power management in DCCs through DVFS, it is essential to consider DCCs' physical security constraints related to job size and service-time latency. These factors directly impact the feasibility of workload reallocation and shifting processes between multiple DCC servers, [8], [9]. For instance, excessively shifting time-sensitive workloads between DCCs may lead to data congestion and latency risks.

To reflect on this issue, Ref. [10] investigated a cloud-based DCCs queuing model using DVFS load-dependent strategies. Ref. [11] analyzed the response characteristics of various time-tolerant (workload) jobs concerning service time and total energy consumption, proposing a dual-timescale collaborative optimization model for multi-DCC systems. Ref. [12] developed an emission-aware stochastic optimization model for microgrid system planning considering batch workload scheduling processes. Ref. [13] extended a data-driven distributionally robust optimization model for low-carbon DCC systems, addressing multi-tasking responses and uncertainties in renewable energy integration. Ref. [7] provided a comprehensive survey of efficient task scheduling and resource allocation methodologies in multi-DCC networks. Ref. [14] proposed a workload-aware robust DVFS technique, Gear-DVFS, to address concurrency challenges in mobile-driven workloads. Ref. [15] proposed a distributed shortest remaining time first (SRTF) scheduling model for DCCs workload reallocation, designed to minimize flow completion time while ensuring flow-size availability. Ref. [16] constructed a machine-learning-based scheduling flow detection and packet-switched optical network (PSON) architecture for intra-DCC connectivity and bath workload allocation. These studies thoroughly examine workload reallocation processes through DVFS across different DCCs. Nevertheless, the impact of flexibility potential provided by these workloads hosted on DVFS-enabled DCCs remains unexplored.

Driven by this motivation, this paper analyses the flexibility offered by DVFS-controlled DCCs and integrates this capability into the scheduling framework. A key aspect of this work is the physical security modeling of workload balancing and reallocation strategies managed by DVFS. Building upon this foundation, additional restrictions such as frequency and voltage regulation limitations through DVFS techniques and time-delay requirements for various workload types are reshaped into corresponding physical constraints. Then, these nonlinear constraints are reshaped into a series of linearized constraints through the McCormick Envelopes method. In contrast to previous studies focusing on workload shifting strategies or energy consumption reduction, this research comprehensively analyzes multiple DCC workload reallocations and DCC server self workload regulation through DVFS.

To summarize, the main contributions of this paper are outlined below: *i)* A set of physical security constraints representing DCCs controlled by DVFS is developed. The proposed constraints are also linearized using McCormick Envelopes to improve computational feasibility in UC problems. *ii)* An interval-based workload reallocation management strategy and DVFS configurations for multiple DCC servers are extended. *iii)* A refined UC is formulated considering additional flexibility potential from DVFS-controlled DCCs through workload reallocation processes.

The remainder of this paper is organized as follows. Section II develops a new set of physical constraints to model DCC workload shifting and local energy consumption requirements. The McCormick Envelope method is also employed to linearize the nonlinear constraints, enabling the estimation of DCC flexibility potential. Section III introduces

an interval-based workload reallocation strategy for multiple DCC servers, which integrates the flexibility potential of DVFS-controlled DCCs while ensuring pre-selected service time latency. Section IV incorporates the flexibility potential of DVFS-controlled DCCs into UC and formulates the proposed scheduled model in detail. Section V presents numerical experiments conducted on benchmark test systems to validate the effectiveness of the proposed model. Finally, Section VI summarizes the key findings and insights from the case studies.

II. PHYSICAL SECURITY CONSTRAINTS FOR DVFS-CONTROLLED DCCs

This section introduces a coordinated scheduling framework designed to integrate the flexible response capabilities produced by DCCs into power system operations. Fig. 1 further details the system architecture and scheduled power management facilitated by DVFS-controlled DCCs.

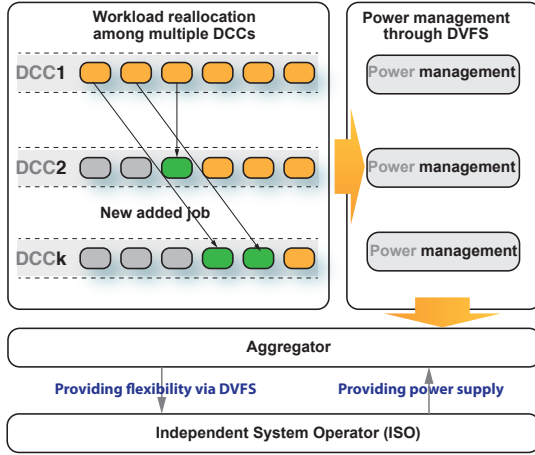


Fig. 1. The scheduled power management through DVFS-controlled DCCs.

The power consumption of DCCs is primarily influenced by two components: static energy and dynamic energy [17]–[20]. The static energy term, often called idle power, arises mainly from the leakage current in the CPU core circuits. In contrast, the dynamic energy term is predominantly determined by the workload ratio λ_{dcc} and the individual service level μ_{dcc} . As such, the total power consumption of DCCs can be expressed as follows, [1], [14],

$$p_{dcc,i,t} = p_{dcc,i,t}^{idle} + p_{dcc,i,t}^{dyn}, \quad \forall i \in \mathcal{I} \quad (1a)$$

$$p_{dcc,i,t}^{idle} = v_{dcc,i,t} i_{leak,i}, \quad \forall i \in \mathcal{I} \quad (1b)$$

$$p_{dcc,i,t}^{dyn} = (C_{dcc,i} f_{dcc,i} v_{dcc,i,t}^2) \lambda_{dcc,i,t} \mu_{dcc,i,t}^{-1}, \quad \forall i \in \mathcal{I} \quad (1c)$$

Based on the physical working principles above, the associated set of security constraints representing DCC workflows can be extended,

$$p_{dcc,i,t} \geq p_{dcc,i,t}^{idle} + p_{dcc,i,t}^{dyn}, \quad \forall i \in \mathcal{I} \quad (2a)$$

$$p_{dcc,i,t}^{idle} \geq [v_{dcc,i,t}] i_{leak,i}, \quad \forall i \in \mathcal{I} \quad (2b)$$

$$p_{dcc,i,t}^{idle} \leq [v_{dcc,i,t}] i_{leak,i}, \quad \forall i \in \mathcal{I} \quad (2c)$$

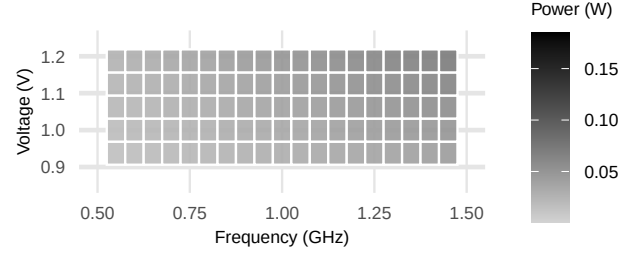


Fig. 2. The impact of power consumption hosted by DCC servers with various frequency/voltage parameters through DVFS techniques.

$$p_{dcc,i,t}^{dyn} \geq [(C_{dcc,i,t} f_{dcc,i} v_{dcc,i,t}^2)] \lambda_{dcc,i,t} \mu_{dcc,i,t}^{-1}, \quad \forall i \in \mathcal{I} \quad (2d)$$

$$p_{dcc,i,t}^{dyn} \leq [(C_{dcc,i,t} f_{dcc,i} v_{dcc,i,t}^2)] \lambda_{dcc,i,t} \mu_{dcc,i,t}^{-1}, \quad \forall i \in \mathcal{I} \quad (2e)$$

where $\lceil \cdot \rceil$ and $\lfloor \cdot \rfloor$ represent predetermined upper and lower bounds for the decision variable $\circ$. The feasibility of workload balancing for compute-intensive and storage-intensive workloads hinges on users' latency tolerance for batch workloads and the capacity to shift workloads among multi-DCCs, [19], [21]. When users accept and tolerate higher latency, the execution of workloads can be dynamically adjusted across multi-DCCs. By optimizing these workload execution patterns, the overall energy consumption of DCCs can be reduced, thereby enhancing their demand response capabilities for power system operations. More detailed, Fig. 2 illustrates the scheduled workload consumption generated by DCC servers with different support voltage and frequency parameter settings.

To estimate the benefits of DVFS-controlled DCCs in achieving smooth energy demand and proactive workload response, additional concise constraints regarding workload balancing processes among multiple DCCs are established,

$$p_{dcc,i,t} \geq p_{dcc,i,t}^{idle} + p_{dcc,i,t}^{dyn} + \Delta p_{dcc,i,t}^{bal}, \quad \forall i \in \mathcal{I} \quad (3a)$$

$$\Delta p_{dcc,i,t}^{bal} = \phi_{dcc,i,t}^{dvfs} (\lambda_{dcc,i,t}^{ini} + \Delta \lambda_{dcc,i,t}) \mu_{dcc,i,t}^{-1}, \quad \forall i \in \mathcal{I} \quad (3b)$$

$$\phi_{dcc,i,t}^{dvfs} = C_{dcc,i,t} f_{dcc,i,t} v_{dcc,i,t}^2, \quad \forall i \in \mathcal{I} \quad (3c)$$

$$[f_{dcc,i}] \leq f_{dcc,i,t} \leq [f_{dcc,i}], \quad \forall i \in \mathcal{I} \quad (3d)$$

$$[v_{dcc,i}] \leq v_{dcc,i,t} \leq [v_{dcc,i}], \quad \forall i \in \mathcal{I} \quad (3e)$$

By incorporating the workload balancing dynamics defined in (3a)–(3c) into (2b) and (2c), the physical security constraints of DVFS-controlled DCC can be established.

In (3a), $\lambda_{dcc,i,t}^{bal}$ primarily depends on the risk of service time latency during workload reallocation processes. Generally, robust tolerance allows for a higher $\lambda_{dcc,i,t}^{bal}$, thereby enhancing the workload balancing capability of DCCs and enabling more flexible adjustment of workloads across DCCs. Nevertheless, an excessively long interval for workload reallocation would lead to prolonged service delays. Specifically, high-resource jobs continuously occupy specific DCC servers while smaller jobs pile up on others servers, [22]. It may lead to a decline in overall resource utilization and a subsequent increase in service time latency. Fig. 3 compares the service-time latency of workload reallocation with different intervals. To mitigate this, the metric of average workload queuing delay duration, i.e., $T_{q,t} = 1/(v_{dcc,i,t} - \lambda_{dcc,i,t})$ is introduced to estimate the performance of workload balancing processes, [20], [23]. The

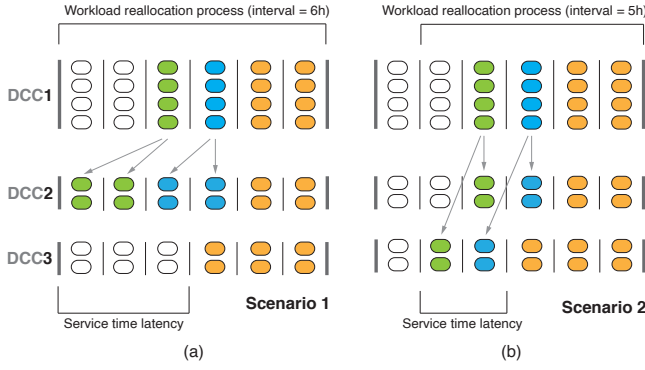


Fig. 3. The variability in service time latency during workload reallocation with different interval configurations.

following outlines the related binding constraints concerning the index $T_{q,t}$,

$$T_{q,t} \leq T_{tol}^{-1} \Rightarrow \quad (4a)$$

$$\Delta\lambda_{dcc,i,t}^{bal} \leq v_{dcc,i,t} - T_{tol}^{-1} - (\lambda_{dcc,i,t}^{ini} + \lambda_{dcc,i,t}^{dyn}), \forall i \in \mathcal{I} \quad (4b)$$

$$\Delta\lambda_{dcc,i,t}^{bal} \geq 0, \Delta\lambda_{dcc,i,t}^{bal} \in \mathbb{R}, \quad \forall i \in \mathcal{I} \quad (4c)$$

Notably, (3a)-(3c) remain intractable due to product terms involving continuous variables. Therefore, the McCormick envelope method, [24], is introduced to linearize them. Let $\iota_{dcc,i,t} = \phi_{dcc,i,t}^{dvfs} \Delta\lambda_{dcc,i,t}$, and additionally introduce two auxiliary variables $\varsigma_{dcc,i,t}^{up}$, $\varsigma_{dcc,i,t}^{dw}$ where $\varsigma_{dcc,i,t}^{up} = 1/2(\phi_{dcc,i,t}^{dvfs} + \Delta\lambda_{dcc,i,t})$, $\varsigma_{dcc,i,t}^{dw} = 1/2(\phi_{dcc,i,t}^{dvfs} - \Delta\lambda_{dcc,i,t})$, accordingly, (3b) can be reformatted as its approximated linearized form as below,

$$\Delta p_{dcc,i,t}^{bal} = \phi_{dcc,i,t}^{dvfs} \lambda_{dcc,i,t}^{ini} \mu_{dcc,i,t}^{-1} + \iota_{dcc,i,t} \mu_{dcc,i,t}^{-1} \quad \forall i \in \mathcal{I} \quad (5a)$$

$$\iota_{dcc,i,t} = \phi_{dcc,i,t}^{dvfs} \Delta\lambda_{dcc,i,t}, \quad \forall i \in \mathcal{I} \quad (5b)$$

$$\iota_{dcc,i,t} \geq (\varsigma_{dcc,i,t}^{up})^2 - (\varsigma_{dcc,i,t}^{dw})^2, \quad \forall i \in \mathcal{I} \quad (5c)$$

$$\varsigma_{dcc,i,t}^{up} \geq 0, \varsigma_{dcc,i,t}^{up} \in \mathbb{R}, \varsigma_{dcc,i,t}^{dw} \in \mathbb{R}, \quad \forall i \in \mathcal{I} \quad (5d)$$

In (4b), the standard square terms of $\langle \varsigma_{dcc,i,t}^{up}, \varsigma_{dcc,i,t}^{dw} \rangle$ can be further reshaped as a Special-Ordered-Sets-of-type-1 (SOS1) constraint, [25], which mix-integer-programming (MIP) solver could accurately identify and compile them. For simplification, let $\langle \varsigma_{dcc,i,t,z}^{up}, \varsigma_{dcc,i,t,z}^{dw} \rangle$, respectively, denote the piecewise block of related continuous variables $\langle \varsigma_{dcc,i,t}^{up}, \varsigma_{dcc,i,t}^{dw} \rangle$, hence, (4b) can be reformulated as below,

$$\iota_{dcc,i,t} \geq \sum_{z \in \mathcal{Z}} \mathfrak{I}(\varsigma_{dcc,i,t,z}^{up}) \varpi_{dcc,i,t,z}^{up} - \sum_{z \in \mathcal{Z}} \mathfrak{I}(\varsigma_{dcc,i,t,z}^{dw}) \varpi_{dcc,i,t,z}^{dw}, \quad \forall i \in \mathcal{I} \quad (6a)$$

$$\varsigma_{dcc,i,t}^{up} = \sum_{z \in \mathcal{Z}} \varsigma_{dcc,i,t,z}^{up} \varpi_{dcc,i,t,z}^{up}, \quad \forall i \in \mathcal{I} \quad (6b)$$

$$\varsigma_{dcc,i,t}^{dw} = \sum_{z \in \mathcal{Z}} \varsigma_{dcc,i,t,z}^{dw} \varpi_{dcc,i,t,z}^{dw}, \quad \forall i \in \mathcal{I} \quad (6c)$$

$$\sum_{z \in \mathcal{Z}} \varsigma_{dcc,i,t,z}^{up} = 1, \sum_{z \in \mathcal{Z}} \varsigma_{dcc,i,t,z}^{dw} = 1, \quad \forall i \in \mathcal{I} \quad (6d)$$

$$\mathfrak{I}(\varsigma_{dcc,i,t,z}^{up}) = (\varsigma_{dcc,i,t,z}^{up})^2, \forall z \in \mathcal{Z}, \quad \forall i \in \mathcal{I} \quad (6e)$$

$$\mathfrak{I}(\varsigma_{dcc,i,t,z}^{dw}) = (\varsigma_{dcc,i,t,z}^{dw})^2, \forall z \in \mathcal{Z}, \quad \forall i \in \mathcal{I} \quad (6f)$$

$$\varsigma_{dcc,i,t,z}^{up} \geq 0, \varsigma_{dcc,i,t,z}^{dw} \geq 0, \forall z \in \mathcal{Z}, \quad \forall i \in \mathcal{I} \quad (6g)$$

$$\varpi_{dcc,i,t,z}^{up} \in \{0,1\}, \varpi_{dcc,i,t,z}^{dw} \in \{0,1\}, \quad \forall z \in \mathcal{Z} \quad (6h)$$

It can be observed that constraint (3b) can continuously retreat as a product of two continuous variables termed as $\pi_{dcc,i,t}^{dvfs}$ and $\Delta\lambda_{dcc,i,t}$, and it can also be equivalent as a set of SOS1 constraint through the McCormick Envelope method. The detailed mathematical formulations for this approximation are provided in Appendix A. By combining these previously mentioned constraints, a comprehensive set of physical constraints describing workload power consumption can be effectively established.

III. INTERVAL-BASED WORKLOAD BALANCING CONSTRAINTS FORMULATION

As previously discussed, $\lambda_{dcc,i,t}$ serves as an indicator for the workload allocation process, where this process is achieved by regulating the operational frequency and supply voltage parameters of each DCC. Nevertheless, the overall workload must remain constant, even though the specific workload assigned to individual DCCs might be adjusted (increased or decreased). To capture this characteristic, the sum of $\lambda_{dcc,i,t}$ for all workloads hosted on a specific DCC server should be restricted as follows,

$$\begin{aligned} \sum_{i \in \mathcal{I}} (\Delta\lambda_{dcc,i,t} + \lambda_{dcc,i,t}^{ini}) &= \sum_{i \in \mathcal{I}} \lambda_{dcc,i,t}^{ini}, \forall i \in \mathcal{I} \\ \Rightarrow \sum_{i \in \mathcal{I}} \Delta\lambda_{dcc,i,t} &= 0, \Delta\lambda_{dcc,i,t} \in \mathbb{R}, \forall i \in \mathcal{I} \end{aligned} \quad (7a)$$

Given that compute workload types differ across DCCs, and not all workloads can be freely transferable to other DCC servers, where this transferability is closely dependent on the current workload type. Specifically, only compute-intensive and storage-intensive types are generally transferable, while network-intensive types may not be. This necessitates an evaluation of each DCC flexibility potential capability to participate in workload reallocation and the establishment of the following constraints.

$$[\Delta\lambda_{dcc,i}] \leq \Delta\lambda_{dcc,i,t} \leq \lceil \Delta\lambda_{dcc,i} \rceil, \quad \forall i \in \mathcal{I} \quad (8a)$$

where $\lceil \Delta\lambda_{dcc,i} \rceil$, $[\Delta\lambda_{dcc,i}]$ denote the upper/lower bound of flexibility potential during workload balancing processes. For these workload categories that can actively respond to the grid's flexible requirements, it is essential to balance local workload at the current period or adjacent periods rather than transferring them over too long a subsequent time interval. This prerequisite is necessary to avoid excessive user waiting time too much. To reflect this, additional restrictions must be constructed,

$$\begin{aligned} \sum_{t \in \mathcal{T}_k} \sum_{i \in \mathcal{I}} \Delta\lambda_{dcc,i,t} &= 0, \\ \forall i \in \mathcal{I}, t \in \mathcal{T}, k \in \{1, 2, 3, \dots, NK\} \end{aligned} \quad (9a)$$

IV. ENHANCED UC CONSIDERING FLEXIBILITY POTENTIAL FROM DVFS-CONTROLLED DCCS

To depict the flexibility potential provided by DCCs, [26], a refined coordination mechanism between workload balancing and the electrical scheduling process, is designed in this

section. A complete mathematical formulation concerning the proposed scheduling model is extended as below,

$$\sum_{t \in \mathcal{T}} \sum_{g \in \mathcal{G}} [C_{su,g} u_{g,t} + C_{sd,g} v_{g,t}] + \quad (10a)$$

$$\sum_{s \in \mathcal{S}} \sum_{t \in \mathcal{T}} \left[\sum_{g \in \mathcal{G}} \left(\sum_{z \in \{1,2,3\}} \alpha_g^{(z)} p_{g,s,t}^{(z)} + x_{g,t} \beta_g^{(z)} \right) + \right. \\ \left. \sum_{w \in \mathcal{W}} C_{shed,w} \Delta p_{w,s,t} + \sum_{d \in \mathcal{D}} C_{voll,d} \Delta p_{d,s,t} \right]$$

$$s.t., u_{g,t} - v_{g,t} = x_{g,t-1} - x_{g,t}, \quad \forall g \in \mathcal{G}, \forall t \in \mathcal{T} \quad (10b)$$

$$x_{g,t} - x_{g,t-1} \leq u_{g,t}, \quad \forall g \in \mathcal{G}, \forall t \in \mathcal{T} \quad (10c)$$

$$\sum_{m \in \mathcal{M}(\omega_g^{dt}-1,t)} (1 - u_{g,m}) \geq \omega_g^{dt} (u_{g,t} - u_{g,t-1}), \\ \forall g \in \mathcal{G}, \forall t \in \mathcal{T} \quad (10d)$$

$$\sum_{m \in \mathcal{M}(\omega_g^{ut}-1,t)} u_{g,m} \geq \omega_g^{ut} (u_{g,t-1} - u_{g,t}), \\ \forall g \in \mathcal{G}, \forall t \in \mathcal{T} \quad (10e)$$

$$\left(\sum_{d \in \mathcal{D}} p_{d,t,s} - \sum_{d \in \mathcal{D}} \Delta p_{d,t,s} \right) + \sum_{i \in \mathcal{I}_b} p_{add,i,t,s} = \\ \sum_{g \in \mathcal{G}} p_{g,s,t} + \left(\sum_{w \in \mathcal{W}} p_{w,t,s} - \sum_{w \in \mathcal{W}} p_{w,t,s} \right) + \\ \left(\sum_{e \in \mathcal{E}} p_{e,t,s}^{dis} - \sum_{e \in \mathcal{E}} p_{e,t,s}^{ch} \right), \\ \forall g \in \mathcal{G}, \forall w \in \mathcal{W}, \forall e \in \mathcal{E}, \forall t \in \mathcal{T}, \forall s \in \mathcal{S} \quad (10f)$$

$$\left| \sum_{b \in \mathcal{B}_l} \ell_{l,b} \left[\sum_{g \in \mathcal{G}_b} p_{g,t,s} - \sum_{i \in \mathcal{I}_b} p_{add,i,t,s} + \right. \right. \\ \left. \left(\sum_{e \in \mathcal{E}_b} p_{e,t,s}^{dis} - \sum_{e \in \mathcal{E}_b} p_{e,t,s}^{ch} \right) + \right. \\ \left. \left(\sum_{w \in \mathcal{W}_b} p_{w,t,s} - \sum_{w \in \mathcal{W}_b} \Delta p_{w,t,s} \right) - \right. \\ \left. \left. \left(\sum_{d \in \mathcal{D}_b} \Delta p_{d,t,s} - \sum_{d \in \mathcal{D}_b} p_{d,t,s} \right) \right] \right| \leq [p_l], \\ \forall l \in \mathcal{L}, \forall t \in \mathcal{T}, \forall s \in \mathcal{S} \quad (10g)$$

$$p_{g,t+1,s} - p_{g,t,s} \leq x_{g,t} \gamma_g^{up} + (u_{g,t} - x_{g,t}) \gamma_g^{su} \\ \forall g \in \mathcal{G}, \forall t \in \mathcal{T}, \forall s \in \mathcal{S} \quad (10h)$$

$$p_{g,t,s} - p_{g,t+1,s} \leq x_{g,t} \gamma_g^{dw} + (v_{g,t} - x_{g,t}) \gamma_g^{sd} \\ \forall g \in \mathcal{G}, \forall t \in \mathcal{T}, \forall s \in \mathcal{S} \quad (10i)$$

$$\lfloor p_g \rfloor x_{g,t} \leq p_{g,t,s} \leq \lceil p_g \rceil x_{g,t}, \quad \forall g \in \mathcal{G}, \forall t \in \mathcal{T}, \forall s \in \mathcal{S} \quad (10j)$$

$$0 \leq \Delta p_{w,t,s} \leq p_{w,t,s}, \quad \forall w \in \mathcal{W}, \forall t \in \mathcal{T}, \forall s \in \mathcal{S} \quad (10k)$$

$$0 \leq \Delta p_{d,t,s} \leq p_{d,t,s}, \quad \forall d \in \mathcal{D}, \forall t \in \mathcal{T}, \forall s \in \mathcal{S} \quad (10l)$$

$$0 \leq p_{e,t,s}^{ch} \leq \lceil p_e^{ch} \rceil \varepsilon_{e,s,t}^{ch}, \quad \forall e \in \mathcal{E}, \forall t \in \mathcal{T}, \forall s \in \mathcal{S} \quad (10m)$$

$$0 \leq p_{e,t,s}^{dis} \leq \lceil p_e^{dis} \rceil \varepsilon_{e,s,t}^{dis}, \quad \forall e \in \mathcal{E}, \forall t \in \mathcal{T}, \forall s \in \mathcal{S} \quad (10n)$$

$$\varepsilon_{e,t,s}^{ch} + \varepsilon_{e,t,s}^{dis} \leq 1, \quad \forall e \in \mathcal{E}, \forall t \in \mathcal{T}, \forall s \in \mathcal{S} \quad (10o)$$

$$q_{e,t,s} = q_{e,t-1,s} + \sum_{t \in \mathcal{T}} [\eta_e^{ch} p_{e,t,s}^{ch} - (1 - \eta_e^{dis}) p_{e,t,s}^{dis}], \\ \forall e \in \mathcal{E}, \forall t \in \mathcal{T}, \forall s \in \mathcal{S} \quad (10p)$$

$$q_{e,t,s} - q_{e,t-1,s} \leq \gamma_e^{ch} \varepsilon_{e,t,s}^{ch}, \quad \forall e \in \mathcal{E}, \forall t \in \mathcal{T}, \forall s \in \mathcal{S} \quad (10q)$$

$$q_{e,t-1,s} - q_{e,t,s} \leq \gamma_e^{dis} \varepsilon_{e,t,s}^{dis}, \quad \forall e \in \mathcal{E}, \forall t \in \mathcal{T}, \forall s \in \mathcal{S} \quad (10r)$$

$$cons.(2b) - (2e), cons.(3d) - (3e), cons.(5a) - (5c),$$

$$cons.(6a) - (6h), \quad \forall i \in \mathcal{I}, \forall t \in \mathcal{T}, \forall s \in \mathcal{S} \quad (10s)$$

$$con.(7a), and, con.(8a), \quad \forall i \in \mathcal{I}, \forall t \in \mathcal{T}, \forall s \in \mathcal{S} \quad (10t)$$

In the above formulations, The objective function (10a) aims

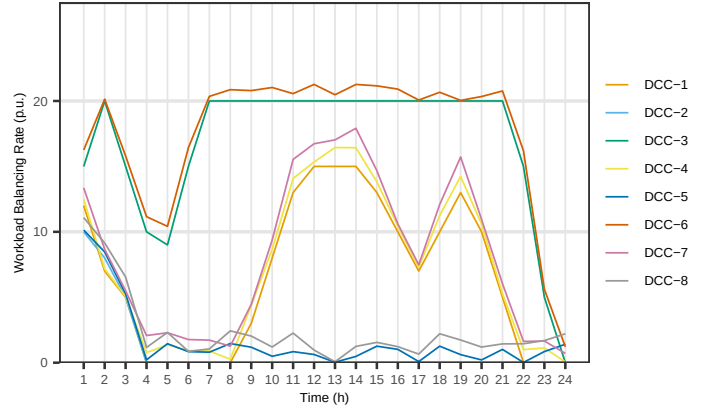


Fig. 4. Results of Workload consumption hosted by multiple DCC servers.

to achieve the economic operation cost of the power systems. Constraints (10b) and (10c) enforce the consistency among unit start-up state, shut-down state, and On/Off status decisions produced by conventional generators. Constraints (10d) and (10e) impose minimum shut-up/shut-down duration restrictions for each conventional generator, ensuring adequate duration between consecutive start-up and shut-down actions. Constraint (10f) ensures a power balance between generation and demand. The demand side includes both conventional electricity consumers and DCCs. Constraint (10g) describes the power flow limits of each transmission line. Constraints (10h) enforce the upward and downward ramping limitations of individual conventional generators. Constraints (10i) and (10j) specify upper/lower bounds of power output produced by each conventional generator. Constraint (10k) restricts the allowable curtailment of wind power, while constraint (10l) accounts for potential forced load curtailment during specific periods. Constraints (10m) and (10n) govern the charging/discharging power of BESS. Constraint (10o) ensures mutual exclusivity between BESS's charging and discharging processes, preventing simultaneous charging and discharging processes. Constraint (10p) tracks the state of charge in BESS, where η_e^{ch} and η_e^{dis} denote the charging and discharging efficiencies, respectively. Constraints (10q) and (10r) define the ramping limits for battery systems during charging and discharging. Constraints (10s) and (10t) represent the physical operating limits of DVFS-controlled DCCs during workload reallocation processes and DVFS regulation processes.

V. CASE STUDIES AND ANALYSIS

To validate the effectiveness of the proposed refined UC model, a data center system comprising 8 DCC servers is integrated into the IEEE 6-bus test system, [27]. The total workload consumption throughout the scheduling horizon is described in Fig. 4. Additionally, two wind farms, each with an installed capacity of 5 MW, are connected at bus 2 in the power system. The sequential load curves and stochastic wind farm outputs are illustrated in Fig. 5 and Fig. 6, respectively. All experiments are conducted on a personal Lenovo desktop with an Intel(R) Core i7-13700 F CPU @3.46 GHz and RAM 32 GB. The proposed and benchmark models are implemented in the Julia-v1.1.5/JuMP-v1.26, [28], environment.

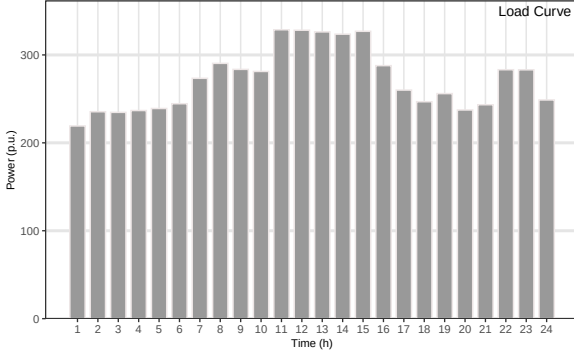


Fig. 5. Chronological load curve of the IEEE 6-bus system.

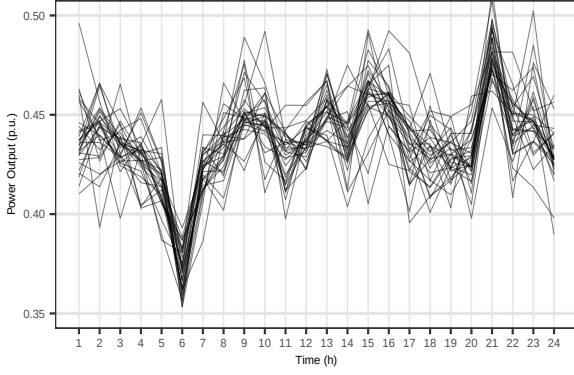


Fig. 6. Chronological stochastic output generated by wind farms.

TABLE I
BOUNDARY PARAMETERS OF DVFS-CONTROLLED DCCs

NO.	Bus	$[p_{dcc,i}]$	$[p_{dcc,i}]$	$\lambda_{dcc,i}^{ini}$	C_{sv}	$\lambda_{dcc,i}^{dyn}$	$\mu_{dcc,i}$
1	1	0.5	0.10	0.05	0.30	0.50	1.0
2	2	0.8	0.16	0.05	0.30	0.75	1.0
3	6	0.4	0.08	0.05	0.30	0.80	1.0
4	2	0.6	0.12	0.05	0.30	0.30	1.0
5	6	0.5	0.10	0.05	0.30	0.40	1.0
6	2	0.7	0.14	0.05	0.30	0.40	1.0
7	2	0.8	0.16	0.05	0.30	0.60	1.0
8	1	0.5	0.10	0.05	0.30	0.20	1.0

TABLE II
THE UPPER/LOWER BOUNDS OF FREQUENCY PARAMETER AND VOLTAGE
PARAMETER GENERATED BY DVFS-CONTROLLED DCC SERVERS.

Parameter type	Lower-bound (p.u.)	Upper-bound (p.u.)
$f_{dcc,i,t}$	0.50	1.50
$v_{dcc,i,t}$	0.85	1.25

The parameters concerning DCCs referring to [12] are configured as follows: The idle workload consumption generated by each DCC server is 10 kW, and 8 DCC servers are equipped to construct all workload requirements. Other boundary conditions consisting of DCC server peak demand, workload ratio, initial frequency and supply voltage, DVFS regulation regions, etc., are detailed in Table. I. The adjustable ranges for DCC frequency parameter and voltage parameters during DVFS processes are [0.50 p.u., 1.50 p.u.], and [0.85 p.u., 1.25 p.u.], respectively. Table. II summarizes the upper/lower bounds of $f_{dcc,i,t}$ and $v_{dcc,i,t}$ accordingly.

To investigate the influence of DCCs in providing flexibility during the entire scheduling process, two models are

TABLE III
SCHEDULING RESULTS OF DIFFERENT UC MODELS WITH/WITHOUT
CONSIDERING FLEXIBILITY POTENTIAL FROM WORKLOAD RESPONSE.

Method	On / Off cost ($\times 10^4$ \$)	Operation cost ($\times 10^4$ \$)	Total cost ($\times 10^4$ \$)
Benchmark	1.17	15.23	16.40
The proposed model	0.91	10.27	11.18

employed: *i*) the traditional UC model that considers DCCs consumption (benchmark), the workload generated by DCC servers is retreated as a standard electrical demand. The comprehensive formulation of this model is available in [6]. *ii*) the proposed UC model considering flexible characteristic generated by DCCs, where the flexibility potential of DCCs is analyzed through (2b) – (2e), (3d) – (3e), (5a) – (5c), and (6a) – (6h). The detailed formulation of this proposed model can be found in (10a) – (10t). A comparative analysis of the scheduling results from both UC models is presented in Fig. III. Additionally, Fig. 7 illustrates the power balance between activated generators and demand, while also taking DVFS-controlled DCCs into account. Furthermore, Fig. 10 depicts the stacked workload consumption using DVFS techniques.

Table III demonstrates that the scheduling costs from the proposed UC model are lower than those from the traditional UC model. This improvement stems from the additional flexible adjustment capabilities provided by DCCs through DVFS techniques in the proposed UC framework. Compared to the benchmark model, the proposed UC model reduces the decorating operations of thermal generators, which are typically caused by their frequent tracking of load fluctuations and wind power uncertainties. This allows thermal generators to maintain a more economical operating state. Thus, the calculated scheduling results show that the proposed model is more economical. This highlights the critical role of DCC flexibility in enhancing both system economics and operational adjustable capabilities.

Fig. 8 shows the workload reallocation and corresponding voltage and frequency adjustments for balancing loads across multiple DCC servers. Significant workload transfers occur during periods 4h and 13h, where the output from wind farms and thermal generators is sufficient. Specially, the workload is increased on lightly loaded servers (DCC1, DCC2, DCC4, DCC5, DCC7, and DCC8) at period 4h. This preemptively reduces the workload on heavily loaded servers (DCC3 and DCC6) in the subsequent periods 5h and 6h. This reallocation allows thermal generators to operate more smoothly and maintain a higher but more economical operation state, facilitating economic efficiency improvement and reducing the need for system flexibility under periods 5h-6h.

Besides, the scheduled workload fluctuations exhibit significant synchronous (see Fig. 8). By taking DCC7 and DCC8 as a case, both workload rates of these two DCC servers keep a synchronized upward trend during periods 1h-4h, while both remain relatively stable during 5h-8h intervals. This consistent phenomenon is primarily attributed to the predefined service-time constraints denoted by (9a). Furthermore, constraint (9a) denotes a functional but straightforward interval-

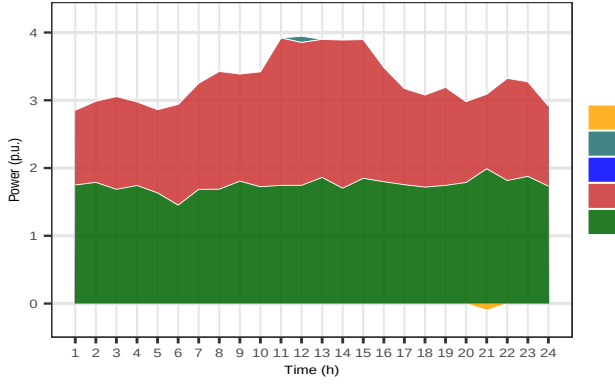


Fig. 7. Power balance process of the IEEE 6-bus system considering workload consumptions hosted by DCCs.

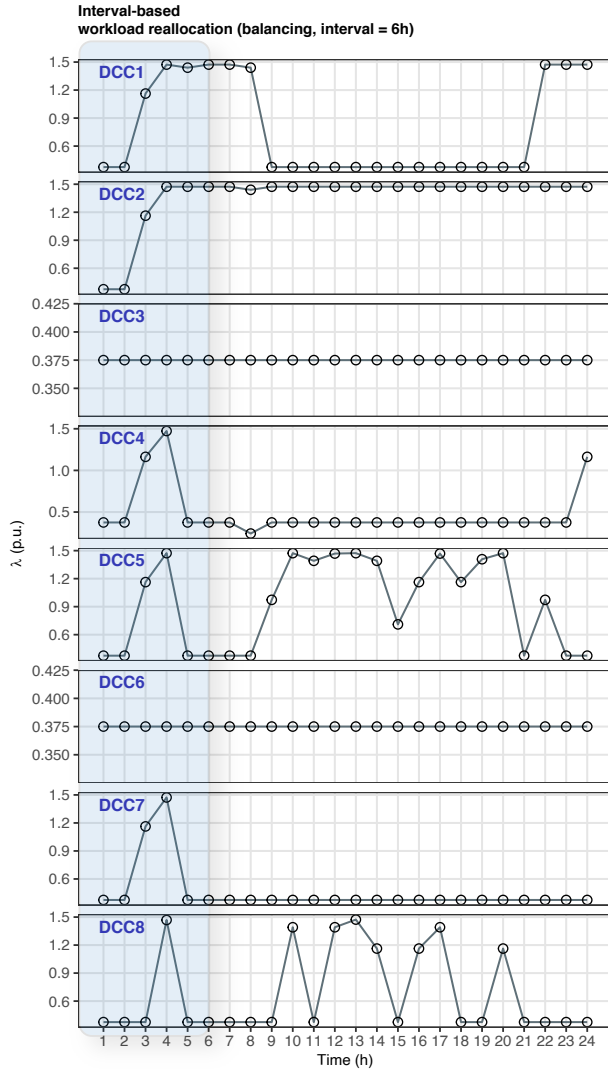


Fig. 8. Results of interval-based workload reallocation across multiple DCCs servers (interval = 6h).

based workload allocation and division strategy. Using this strategy, the entire scheduling horizon (24h) is segmented into a series of independent intervals such as 0h-6h, 7h-12h, 13h-18h, 19h-24h. Under each independent scheduling interval, the

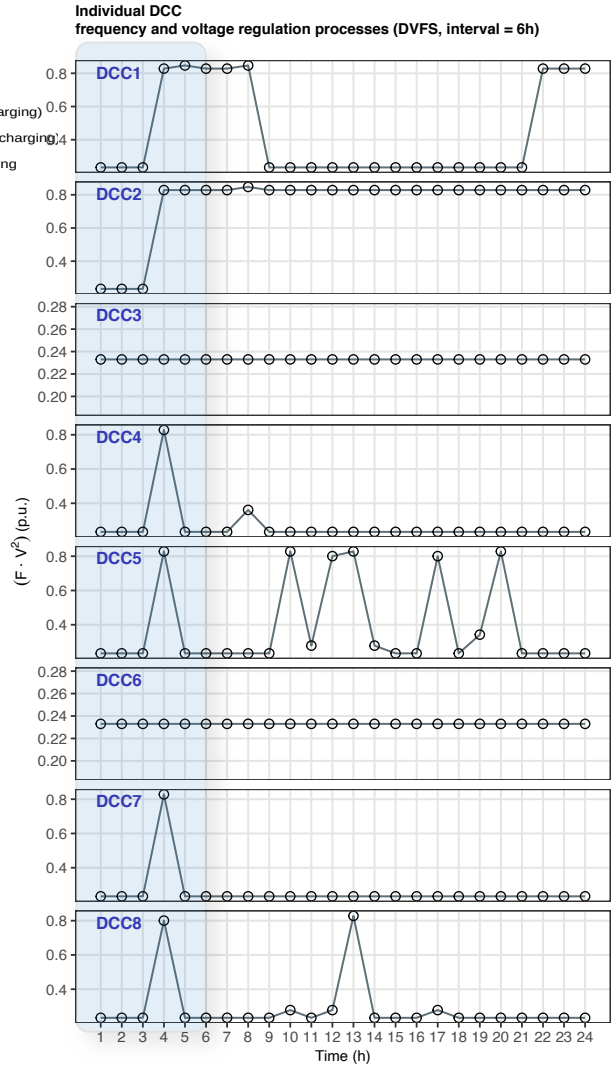


Fig. 9. Results of voltage and frequency regulation parameters for individual DCC servers using the DVFS technique.

workload reallocation across multiple DCCs is permitted to achieve workload balancing optimization. Conversely, cross-interval workload reallocation between adjacent scheduling intervals would be prohibited to maintain system efficiency and minimize inter-dependencies. Hence, the system would leverage workload flexibility with maximum efficiency and ensure the minimization of service times by confining workload balancing operations to designated time intervals.

Moreover, a similar synchronous adjustment characteristic in $f_{dcc,I,t} v_{dcc,I,t}^2$ in Fig. 9 can also be observed. Specifically, all DCC servers other than DCC3 / DCC6 simultaneously initiated their CPU frequency parameter and voltage parameters through DVFS mechanisms, exhibiting a highly consistent adjustment trend. Similarly, DCC5 and DCC8 also implemented DVFS regulation synchronously during the period 12h. This phenomenon indicates a close synergistic relationship between workload-balancing processes among DCCs and the intrinsic DVFS processes. Both workload reallocations (across multiple DCCs) and instant frequency/voltage parameter regulations (hosted on individual DCC servers) maintain synchronicity in

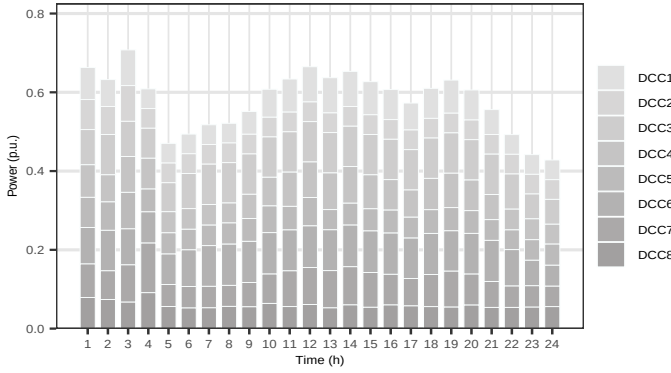


Fig. 10. Sequential workload consumption generated by multiple DCCs reallocation (intervals = 6h).

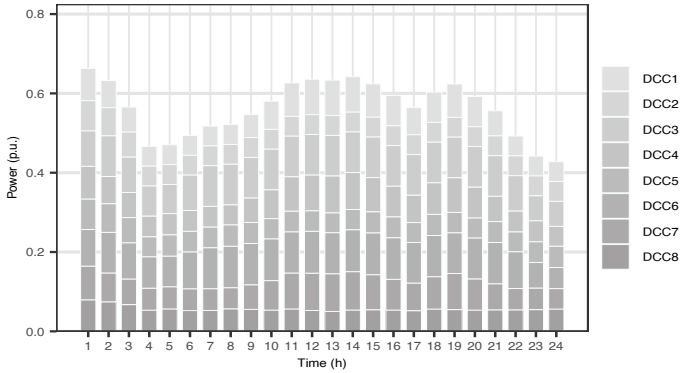


Fig. 11. Sequential workload consumption generated by multiple DCCs reallocation (intervals = 4h).

the temporal dimension and keep unidirectional adjustment in their regulation directions.

Despite observing synchronous trends, the fluctuations between $f_{dcc,i,t}v_{dcc,i,t}^2$ (in Fig. 9) and $\lambda_{dcc,i,t}$ (in Fig. 8) are not entirely identical. The difference in this discrepancy lies in the varying load-rate levels of various DCCs and possible adjustable difficulties in migrating workload from heavily loaded DCC servers to lightly loaded ones. This disparity is reflected in the distinct distributions of the $\lambda_{dcc,i,t}$ curve and the $f_{dcc,i,t}v_{dcc,i,t}^2$ curve.

To comprehensively compare workload consumption and flexibility potential across different balancing intervals, Fig. 11 and Fig. 10 illustrate the sequential workload consumption after multiple DCC reallocation during 4h and 6h intervals, respectively in comparison with the original workload consumption amount 14.93 p.u. Without workload-balancing management, the reallocated workload consumption through multiple DCCs balancing processes and DVFS processes is 13.94 MW, where the reduction of workload consumption is about 0.99 MW. When the interval-based workload reallocation period is extended to 6h, the overall workload consumption requirement drops to 13.51MW, a decrease of 1.42MW compared to the unbalanced case. The reduction of workload consumption through multiple DVFS-controlled DCC balancing processes directly demonstrates the validity of this work. It can be concluded that appropriately lengthening the workload balancing

interval leads to better energy optimization and reduced total workload consumption, thereby improving the power system's operational economy. However, it is crucial to acknowledge that excessively prolonging the pre-elected interval for workload reallocation will increase service latency. Therefore, considering the trade-offs between workload consumption and service latency is still essential.

VI. CONCLUSION

This paper proposes a refined UC model that incorporates the flexibility capabilities of DCCs through DVFS techniques. Specifically, the flexibility potential gained from reallocating workloads across multiple DCC servers and the real-time adjustments of frequency and voltage parameters for individual DCC servers is investigated. Building on this, a set of physical security constraints representing these processes is extended. Furthermore, these constraints are reshaped as SOS1-derived linearized constraints and subsequently integrated into the scheduling framework.

Several findings in case studies suggested that the flexibility potential of workloads hosted by DVFS-controlled DCCs could be effectively exploited by adjusting the interval duration during which DCCs participate in workload reallocation. This work can reduce the overall operation cost of power systems without excessively delaying the service latency of workload response.

VII. APPENDIX A: REFORMULATION OF CONSTRAINTS (3C) THROUGH MCCORMICK ENVELOPE METHOD

Let $\gamma_{dcc,i}$ denotes the square value of $v_{dcc,i}^2$, i.e., $\gamma_{dcc,i} = v_{dcc,i}^2$, the lower/upper bound of $\gamma_{dcc,i}$ can be obtained through estimating bindings parameters $(\lfloor v_{dcc,i} \rfloor)^2$ and $(\lceil v_{dcc,i} \rceil)^2$, respectively. This process holds because of the monotonic increasing property of the quadratic function. Accordingly, the constraints representing the lower/upper bounds of $\gamma_{dcc,i}$ can be derived as below,

$$\lfloor v_{dcc,i} \rfloor \leq v_{dcc,i} \leq \lceil v_{dcc,i} \rceil, \quad \forall i \in \mathcal{I} \quad (11a)$$

$$\Rightarrow (\lfloor v_{dcc,i} \rfloor)^2 \leq \gamma_{dcc,i} \leq (\lceil v_{dcc,i} \rceil)^2, \quad \forall i \in \mathcal{I} \quad (11b)$$

Next, $\phi_{dcc,i}^{dvfs}$ can be equivalently represented as the product of two continuous variables $\gamma_{dcc,i}$ and $f_{dcc,i}$. By utilizing the McCormick Envelope method, a reformulated, linearized approximation of this product, and thus of $\phi_{dcc,i}^{dvfs}$ can be constructed as,

$$\kappa_{dcc,i}^{up} = 1/2(\gamma_{dcc,i} + f_{dcc,i}), \quad \forall i \in \mathcal{I} \quad (12a)$$

$$\kappa_{dcc,i}^{dw} = 1/2(\gamma_{dcc,i} - f_{dcc,i}), \quad \forall i \in \mathcal{I} \quad (12b)$$

$$\phi_{dcc,i}^{dvfs} \geq \sum_{z \in \mathcal{Z}} \Im(\kappa_{dcc,i,z}^{up}) \vartheta_{dcc,i,z}^{up} - \sum_{z \in \mathcal{Z}} \Im(\kappa_{dcc,i,z}^{dw}) \vartheta_{dcc,i,z}^{dw}, \quad \forall i \in \mathcal{I} \quad (12c)$$

$$\kappa_{dcc,i}^{up} = \sum_{z \in \mathcal{Z}} \kappa_{dcc,i,z}^{up} \vartheta_{dcc,i,z}^{up}, \quad \forall i \in \mathcal{I} \quad (12d)$$

$$\kappa_{dcc,i}^{dw} = \sum_{z \in \mathcal{Z}} \kappa_{dcc,i,z}^{dw} \vartheta_{dcc,i,z}^{dw}, \quad \forall i \in \mathcal{I} \quad (12e)$$

$$\sum_{z \in \mathcal{Z}} \vartheta_{dcc,i,z}^{up} = 1, \quad \forall i \in \mathcal{I} \quad (12f)$$

$$\sum_{z \in \mathcal{Z}} \vartheta_{dcc,i,z}^{dw} = 1, \quad \forall i \in \mathcal{I} \quad (12g)$$

$$\sum_{z \in \mathcal{Z}} \vartheta_{dcc,i,z}^{dw} = 1, \quad \forall i \in \mathcal{I} \quad (12h)$$

$$\mathfrak{S}(\kappa_{dcc,i,z}^{up}) = (\kappa_{dcc,i,z}^{up})^2, \quad \forall i \in \mathcal{I} \quad (12i)$$

$$\mathfrak{S}(\kappa_{dcc,i,z}^{dw}) = (\kappa_{dcc,i,z}^{dw})^2, \quad \forall i \in \mathcal{I} \quad (12j)$$

$$\kappa_{dcc,i,z}^{up} \geq 0, \kappa_{dcc,i,z}^{dw} \geq 0, \quad \forall z \in \mathcal{Z}, \quad \forall i \in \mathcal{I} \quad (12k)$$

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